Geometry Review: Unit 8 — Coordinate Geometry & Transformations

Station 1 of 4: Distance, Midpoint, and Slope in the Coordinate Plane

Name: _____ Date: _____

Give exact answers in simplest form. Round decimals to the nearest tenth only when asked. Diagrams are not drawn to scale.

1. Find the distance between $A(1, 2)$ and $B(4, 6)$.

2. Find the distance between $A(-3, -1)$ and $B(2, 11)$.

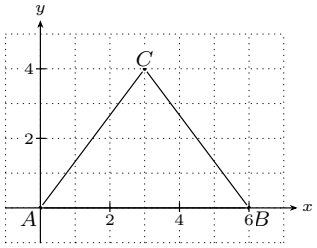
3. Find the distance between $(0, 0)$ and $(5, 5)$. Give the exact answer and a decimal to the nearest tenth.

4. Find the midpoint of the segment with endpoints $(-4, 7)$ and $(6, -3)$.

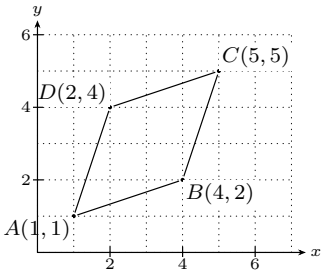
5. $M(2, 5)$ is the midpoint of $\overline{AB}$ and A is $(-1, 3)$. Find B.

6. Find the slope of the line through $A(-2, 3)$ and $B(4, -1)$.

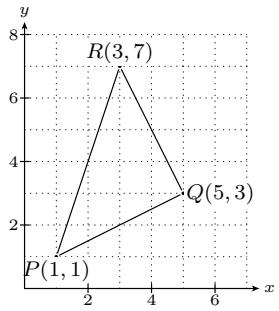
7. A triangle has vertices $A(0, 0)$, $B(6, 0)$, and $C(3, 4)$. Find its perimeter and classify it by its sides.



8. Is the quadrilateral with vertices $A(1, 1)$, $B(4, 2)$, $C(5, 5)$, $D(2, 4)$ a parallelogram? Use slopes.



9. Show that $P(1, 1)$, $Q(5, 3)$, $R(3, 7)$ form a right triangle by finding slopes.



10. Find the point that is $\frac{1}{3}$ of the way from $A(0, 0)$ to $B(9, 12)$.

Geometry Review: Unit 8 — Coordinate Geometry & Transformations

Station 2 of 4: Equations of Circles

Name: _____ Date: _____

Use $(x-h)^2+(y-k)^2=r^2$. Leave radicals in simplest radical form. Diagrams are not drawn to scale.

1. Write the equation of the circle with center $(0,0)$ and radius 5.

2. Write the equation of the circle with center $(3,-2)$ and radius 4.

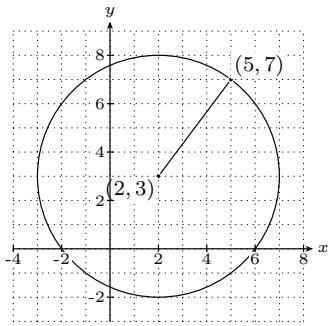
3. Write the equation of the circle with center $(-1,5)$ and radius $\sqrt{7}$.

4. Find the center and radius of $(x-4)^2+(y+1)^2=36$.

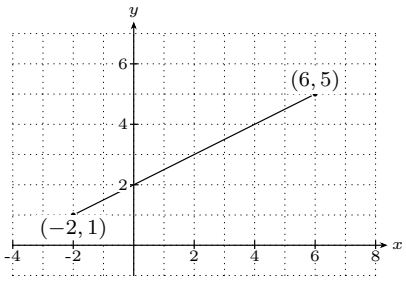
5. Find the center and radius of $x^2+y^2=49$.

6. Find the center and radius of $x^2+(y-3)^2=20$.

7. A circle has center $(2,3)$ and passes through $(5,7)$. Write its equation.



8. A circle has a diameter with endpoints $(-2,1)$ and $(6,5)$. Write its equation.



9. For the circle $x^2+y^2=25$: is $(3,4)$ on the circle? Is $(2,5)$ inside, on, or outside the circle?

10. Write $x^2+y^2-6x+4y-12=0$ in standard form and identify the center and radius.

Geometry Review: Unit 8 — Coordinate Geometry & Transformations

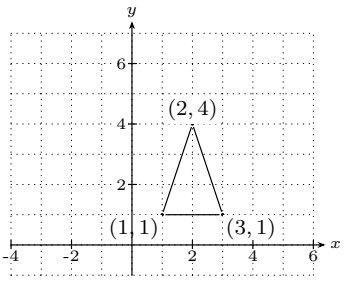
Station 3 of 4: Translations, Reflections, and Rotations

Name: _____ Date: _____

Find the coordinates of each image. Rotations are about the origin unless stated. Diagrams are not drawn to scale.

1. Translate $A(2, 3)$ by $(x, y) \to (x + 4,\ y - 5)$.

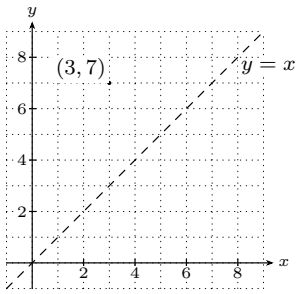
2. Translate the triangle $(1, 1)$, $(3, 1)$, $(2, 4)$ by $(x, y) \to (x - 3,\ y + 2)$.



3. Reflect $(4, -3)$ across the x-axis.

4. Reflect $(-5, 2)$ across the y-axis.

5. Reflect $(3, 7)$ across the line $y = x$.



6. Reflect $(2, 5)$ across the line $x = 4$.

7. Rotate $(3, 1)$ by $90°$ counterclockwise about the origin.

8. Rotate $(-2, 5)$ by $180°$ about the origin.

9. Rotate $(4, -2)$ by $90°$ clockwise about the origin.

10. Rotate $(6, -1)$ by $270°$ counterclockwise about the origin.

Geometry Review: Unit 8 — Coordinate Geometry & Transformations

Station 4 of 4: Dilations and Compositions of Transformations

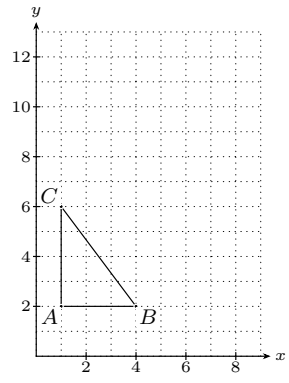
Name: _____ Date: _____

Dilations are centered at the origin unless stated otherwise. Diagrams are not drawn to scale.

1. Dilate $(2, -3)$ by a scale factor of 3.

2. Dilate $(8, 12)$ by a scale factor of $\frac{1}{4}$.

3. Dilate triangle $A(1, 2)$, $B(4, 2)$, $C(1, 6)$ by a scale factor of 2.



4. A dilation maps $A(2, 4)$ to $A'(6, 12)$. Find the scale factor.

5. Dilate $(-6, 9)$ by a scale factor of $\frac{2}{3}$.

6. A segment of length 5 is dilated by a scale factor of 4. Find the length of the image.

7. Dilate $(3, 4)$ by a scale factor of 2 with center of dilation $(1, 1)$.

8. Translate $(3, 5)$ by $(x, y) \to (x + 2,\ y - 1)$, then reflect the image across the x-axis.

9. Reflect $(-2, 3)$ across the y-axis, then rotate the image $90°$ counterclockwise about the origin.

10. A figure with area 12 is dilated by a scale factor of 3. Find the area of the image.

Geometry Review: Unit 8 — Coordinate Geometry & Transformations

Answer Key

Station 1: Distance, Midpoint, and Slope in the Coordinate Plane

1. 5
2. 13
3. $5\sqrt{2} \approx 7.1$
4. $(1, 2)$
5. $B(5, 7)$
6. $-\dfrac{2}{3}$
7. 16; isosceles $AB = 6,\ AC = BC = 5$
8. Yes; $\overline{AB} \parallel \overline{DC}$ (slope $\frac{1}{3}$) and $\overline{AD} \parallel \overline{BC}$ (slope 3)
9. slope $PQ = \dfrac{1}{2}$, slope $QR = -2$; slopes are opposite reciprocals, so the angle at Q is a right angle
10. $(3, 4)$

Station 2: Equations of Circles

1. $x^2 + y^2 = 25$
2. $(x - 3)^2 + (y + 2)^2 = 16$
3. $(x + 1)^2 + (y - 5)^2 = 7$
4. center $(4, -1)$; $r = 6$
5. center $(0, 0)$; $r = 7$
6. center $(0, 3)$; $r = 2\sqrt{5}$
7. $(x - 2)^2 + (y - 3)^2 = 25$ $\quad r = \sqrt{3^2 + 4^2} = 5$
8. $(x-2)^2+(y-3)^2 = 20$ $\quad$ *center* $(2, 3)$*;* $r^2 = 4^2+2^2 = 20$
9. Yes, it is on the circle; $2^2 + 5^2 = 29 > 25$, so outside
10. $(x - 3)^2 + (y + 2)^2 = 25$; center $(3, -2)$, $r = 5$

Station 3: Translations, Reflections, and Rotations

1. $A'(6, -2)$
2. $(-2, 3),\ (0, 3),\ (-1, 6)$
3. $(4, 3)$
4. $(5, 2)$
5. $(7, 3)$
6. $(6, 5)$
7. $(-1, 3)$ $\quad (x, y) \to (-y, x)$
8. $(2, -5)$ $\quad (x, y) \to (-x, -y)$
9. $(-2, -4)$ $\quad (x, y) \to (y, -x)$
10. $(-1, -6)$ $\quad$ *same as* $90°$ *clockwise*

Station 4: Dilations and Compositions of Transformations

1. $(6, -9)$
2. $(2, 3)$
3. $A'(2, 4),\ B'(8, 4),\ C'(2, 12)$
4. 3
5. $(-4, 6)$
6. 20
7. $(5, 7)$ $\quad$ *image* $= (1 + 2 \cdot 2,\ 1 + 2 \cdot 3)$
8. $(5, -4)$
9. $(-3, 2)$ $\quad (2, 3) \to (-3, 2)$
10. 108 $\quad 12 \cdot 3^2$